FY2023 financial statements, unsupervised profiles, and supervised distress prediction

Problem and Data

Objective

- Profile SMEs using financial characteristics.
- Predict `Financial distress`.
- Produce interpretable evidence for lenders, policy makers, and firms.

Dataset

- 13,956 Italian SMEs.
- 14 feature columns plus binary target.
- 108 provinces, 306 sectors, 268 ATECO codes.

Target imbalance

Class	Firms
Not distressed	12,451
Distressed	1,505
Distress rate	10.78%

Because the positive class is rare, model quality is judged with F1, balanced accuracy, recall, precision, ROC-AUC, and average precision.

Clustering Design

Principle

- Cluster on financial behavior only.
- Exclude `Financial distress`, `Province`, `sector`, and `Ateco`.
- Use geography and sector only after clustering, for interpretation.

Preprocessing

- Signed logs for skewed monetary variables.
- Margins and per-employee productivity ratios.
- Debt-pressure and cash-flow ratios.
- Winsorization at 0.5th and 99.5th percentiles.
- Median imputation and robust scaling.

Chosen solution

$$k = 5$$

A financially interpretable ladder:

1. resilient cash generators
2. healthy profitable firms
3. thin-margin vulnerable firms
4. debt-burdened break-even firms
5. loss-making cash-flow distressed firms

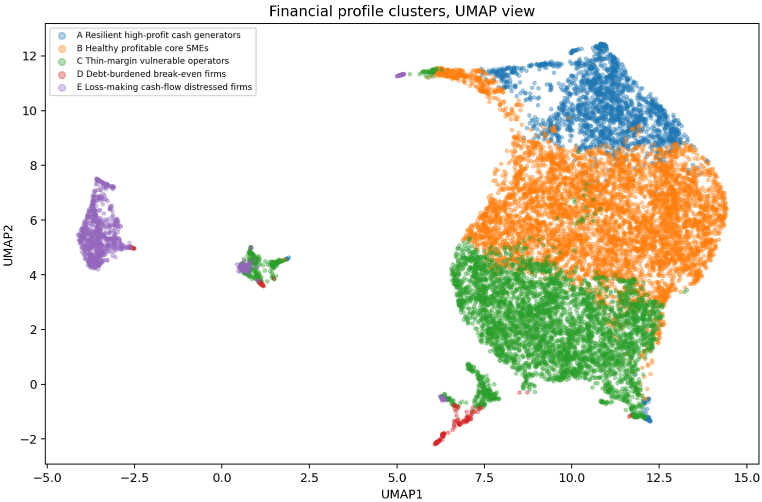
The k=5 solution is more useful for presentation than the trivial two-cluster split between broadly healthy and broadly distressed firms.

Five Financial Profiles

Cluster	Firms	Distress	Net margin	OCF margin	Interpretation
A Resilient high-profit cash generators	2,085	0.43%	15.6%	19.7%	strong profits, strong cash flow, very low financial-expense pressure
B Healthy profitable core SMEs	6,043	2.75%	4.5%	7.4%	stable middle: profitable, cash-generative, moderate scale
C Thin-margin vulnerable operators	4,745	15.68%	0.7%	3.2%	low margin, lower productivity, more sensitive to shocks
D Debt-burdened break-even firms	174	31.61%	-1.8%	1.5%	near break-even operations with high financial-expense burden
E Loss-making cash-flow distressed firms	909	58.42%	-10.2%	-6.1%	losses and negative operating cash flow dominate

Main interpretation: distress rises smoothly as profitability, productivity, and operating cash flow deteriorate.

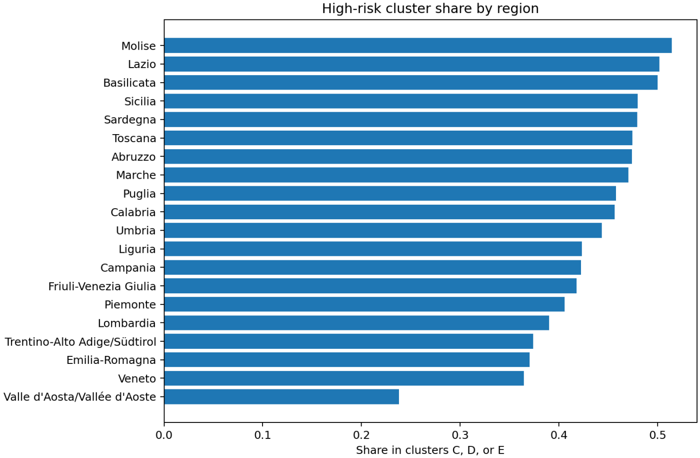
Cluster Map: UMAP View



Why this plot matters

- UMAP projects the same financial feature space used for k-means.
- The loss-making profile separates clearly.
- Thin-margin and healthy firms form the large operating core.
- The small debt-burdened group appears as a narrow transition profile.

Geographic Pattern



High-risk profiles

- C: thin-margin vulnerable operators
- D: debt-burdened break-even firms
- E: loss-making cash-flow distressed firms

Top shares

Region	Share
Molise	51.43%
Lazio	50.17%
Basilicata	50.00%
Sicilia	48.01%
Sardegna	47.96%

The chart describes concentration of fragile financial profiles, not a causal effect of location.

Prediction Model: Rank Ensemble

Models

- ExtraTrees
- LightGBM
- XGBoost
- HistGradientBoosting
- CatBoost with native categorical features

Ensembling

- Convert each model's OOF scores to ranks.
- Average ranks across models.
- Threshold the rank ensemble.

Out-of-fold performance

Metric	Value
ROC-AUC	0.880
Average precision	0.522
Best-F1 threshold	0.88
F1	0.517
Precision	0.512
Recall	0.522
Accuracy	0.895

If the metric rewards recall or balanced accuracy, the threshold should be lowered to roughly 0.70–0.71.

OOF Scores: True Class vs Ensemble Risk

At threshold 0.88

Outcome	Count
True negatives	11,702
False positives	749
False negatives	719
True positives	786

Distressed firms receive much higher ensemble risk scores on average, but there is overlap near the decision boundary.

What Drives Predicted Distress?

Rank	Feature
1	net income per employee
2	sector
3	cash flow – net income gap
4	taxes / operating income
5	ATECO code
6	financial expenses / abs. operating income
7	revenue per employee
8	province
9	net income signed log
10	operating margin

Interpretation

- Distress is primarily a profitability and cash-flow problem.
- Sector and ATECO add activity-specific risk.
- Province contributes additional geographic signal.
- Debt pressure matters relative to operating capacity.

Final Takeaway

The clustering and prediction stories agree

Firms move from low-risk to high-risk profiles as margins, cash flow, productivity, and debt-service capacity weaken.

Interpretation

- Five clusters provide an economic map of SME vulnerability.
- Geography and sector explain where fragile profiles are concentrated.

Prediction

- Rank ensemble provides robust OOF risk ordering.
- F1 threshold is the default submission choice unless the hidden metric rewards recall more strongly.

The model is useful because it is both predictive and explainable.